

PGDIT, IIT, DU

Assignment: Object Oriented Programming

Simple file processing in Python involves fundamental operations such as reading, writing, and appending text. Please complete the following tasks related to file processing

1. Basic Read/Write

Task: Create a script that creates a file named hello.txt, writes "Hello, Python!" into it, and then reads it back to display the content on the console.

Key Concept: Use open() with mode 'w' for writing and 'r' for reading.

2. Line-by-Line Processing

Task: Read a multi-line text file (e.g., data.txt) and print each line prefixed with its line number (e.g., "1: First line content").

Key Concept: Use a for loop to iterate through the file object and the enumerate() function for the index.

3. File Statistics Counter

Task: Write a function that reads a file and returns a dictionary containing the total number of lines, words, and characters.

Key Concept: Combine file reading with string methods like .split() to count words and len() for characters.

4. Search and Replace

Task: Open an existing text file, find all occurrences of a specific word (e.g., "fox"), replace them with another (e.g., "cat"), and save the result into a new file.

Key Concept: Use the string.replace() method on the file content before writing it to a new destination.

5. Data Filtering

Task: Read a text file containing one integer per line. Filter these numbers and write only the even integers into a separate file called even_numbers.txt.

Key Concept: Practice conditional logic while reading from one file and writing to another.

6. Appending Data

Task: Prompt a user to enter their name and append that name to an existing roster.txt file without deleting the current contents.

Key Concept: Open the file in append mode ('a') to add data to the end of the file.